

TECHNICAL REPORT · 2026

CheXNet

Radiologist-Level Chest X-Ray Classification

PyTorch implementation of the Stanford CheXNet paper.

Multi-label classification of 14 thoracic diseases
from chest X-rays using DenseNet-121.

Dataset	Backbone	Classes
NIH ChestX-ray14	DenseNet-121	14 Pathologies

Based on: Rajpurkar et al. [arXiv:1711.05225](#)

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1. Introduction

Chest X-ray (CXR) imaging is the most widely used diagnostic tool in clinical radiology, yet its interpretation is time-consuming and highly expertise-dependent. In 2017, Rajpurkar et al. from the Stanford Machine Learning Group published **CheXNet**, demonstrating that a convolutional neural network could detect pneumonia from chest X-rays at a level exceeding the average performance of four practicing radiologists.

This report documents a faithful PyTorch reimplementation of CheXNet, trained on the **NIH ChestX-ray14** dataset. The implementation covers the full pipeline: dataset preparation, model architecture, training with checkpoint resumption, per-class AUC-ROC evaluation, and Grad-CAM explainability.



Paper Reference

CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning

Rajpurkar P., Irvin J., Zhu K., Yang B., Mehta H., Duan T., Ding D., Bagul A., Langlotz C., Shpanskaya K. et al. — arXiv:1711.05225, Stanford ML Group, 2017.

2. Dataset

The **NIH ChestX-ray14** dataset contains 112,120 frontal-view chest X-ray images from 30,805 unique patients. Disease labels were extracted from associated radiology reports using NLP techniques, producing weak multi-label annotations across 14 thoracic pathology classes.

Atelectasis	Cardiomegaly	Effusion	Infiltration	Mass	Nodule	Hernia
Pneumonia	Pneumothorax	Consolidation	Edema	Emphysema	Fibrosis	Pleural Thickening

Table 1 — The 14 thoracic pathology classes in NIH ChestX-ray14.

Data Splits

The official NIH **test_list.txt** defines the held-out test set, ensuring no patient overlap between training and evaluation — a critical requirement given that multiple X-rays can belong to the same patient. The remaining samples are split into train and validation partitions from **Data_Entry_2017.csv**.

Each sample is a **multi-hot binary vector** of length 14, where a value of 1 indicates the presence of the corresponding disease. A single image may be positive for multiple pathologies simultaneously.

Preprocessing & Augmentation

- **Resize** to 256 x 256, then **RandomCrop** to 224 x 224 (train) or center-crop (val/test).
- **RandomRotation** of up to +/-15 degrees and **RandomHorizontalFlip** (p=0.5) during training.
- **Normalization** using ImageNet mean [0.485, 0.456, 0.406] and std [0.229, 0.224, 0.225].

3. Model Architecture

The architecture follows the original CheXNet paper exactly: a **DenseNet-121** backbone pretrained on ImageNet, with the final classification head replaced by a single fully-connected layer projecting the 1024-dimensional feature vector to 14 output logits — one per pathology.

DenseNet-121 Backbone

DenseNet (Densely Connected Convolutional Network) connects each layer to every subsequent layer in a feed-forward fashion. For a network with L layers, there are $L(L+1)/2$ direct connections. This design encourages feature reuse, reduces vanishing gradients, and achieves strong performance with fewer parameters than comparable ResNet architectures.

The DenseNet-121 variant used here contains **4 Dense Blocks** with transition layers between them, totalling 121 learnable layers. The global average pooling output of dimension 1024 feeds directly into the classification head.

Classification Head & Loss

- **Linear(1024 → 14)**: one logit per disease class.
- **BCEWithLogitsLoss**: numerically stable binary cross-entropy applied independently per class, enabling true multi-label learning.
- **Sigmoid** is applied at inference time to convert logits to per-class probabilities in $[0, 1]$.
- **DataParallel** wraps the model to utilise multiple GPUs when available.

Component	Detail
Backbone	DenseNet-121 (ImageNet pretrained)
Input	224 x 224 x 3 RGB
Feature dim	1024-d (global average pool)
Classifier head	Linear(1024, 14)
Output	14 independent probabilities via Sigmoid
Loss	BCEWithLogitsLoss
Multi-GPU	torch.nn.DataParallel

Table 2 — Model architecture summary.

4. Training Procedure

Configuration

Hyperparameter	Value
Input Resolution	224 x 224
Batch Size	32
Optimizer	Adam (epochs 1-13) → AdamW (epochs 14-20)
LR Scheduler	ReduceLROnPlateau (factor=0.1, patience=2)
Loss Function	BCEWithLogitsLoss (multi-label)
Augmentation	RandomCrop, RandomRotation +/-15, RandomHorizontalFlip
Normalization	ImageNet mean / std
Multi-GPU	torch.nn.DataParallel
Checkpointing	Every epoch; best model saved on val-loss improvement

Table 3 — Training hyperparameters and configuration.

Optimizer & Scheduler

Training uses a two-phase optimizer strategy. **Adam** is used for the first 13 epochs to benefit from fast convergence in the early stages. From epoch 14 onward the optimizer switches to **AdamW**, which adds decoupled weight decay regularization and helps prevent overfitting as the model fine-tunes toward convergence. The learning rate is managed throughout by **ReduceLROnPlateau**, which reduces the LR by a factor of 0.1 whenever validation loss fails to improve for 2 consecutive epochs.

Checkpointing & Resume

At the end of every epoch the full training state is serialised to **chexnet_checkpoint.pth**, including model weights, optimizer state, scheduler state, current epoch index, and best validation loss. If this file exists at startup, training resumes seamlessly from the last epoch. The model weights alone are separately saved to **best_model.pth** whenever a new minimum validation loss is achieved.

5. Evaluation & Results

Each of the 14 pathology classes is evaluated independently using the **Area Under the ROC Curve (AUC-ROC)**, consistent with the original paper and standard practice in medical image classification. This metric is threshold-independent and handles the severe class imbalance inherent in the dataset.

At inference, the model outputs a sigmoid probability for each class. These are compared against binary ground-truth labels from the official test split to compute per-class and mean AUC-ROC scores.

Pathology	This Implementation	Original CheXNet	Delta
Atelectasis	0.800	0.809	-0.009
Cardiomegaly	0.923	0.925	-0.002
Effusion	0.853	0.864	-0.011
Infiltration	0.730	0.735	-0.005
Mass	0.872	0.868	+0.004
Nodule	0.797	0.780	+0.017
Pneumonia	0.770	0.768	+0.002
Pneumothorax	0.906	0.889	+0.017
Consolidation	0.790	0.790	0.000
Edema	0.881	0.888	-0.007
Emphysema	0.929	0.937	-0.008
Fibrosis	0.869	0.805	+0.064
Pleural Thickening	0.815	0.806	+0.009
Hernia	0.967	0.916	+0.051
Mean AUC	0.851	0.841	+0.010

Table 4 — Per-class AUC-ROC scores comparing this implementation against the original CheXNet paper results. Green delta (+) indicates improvement over the baseline.

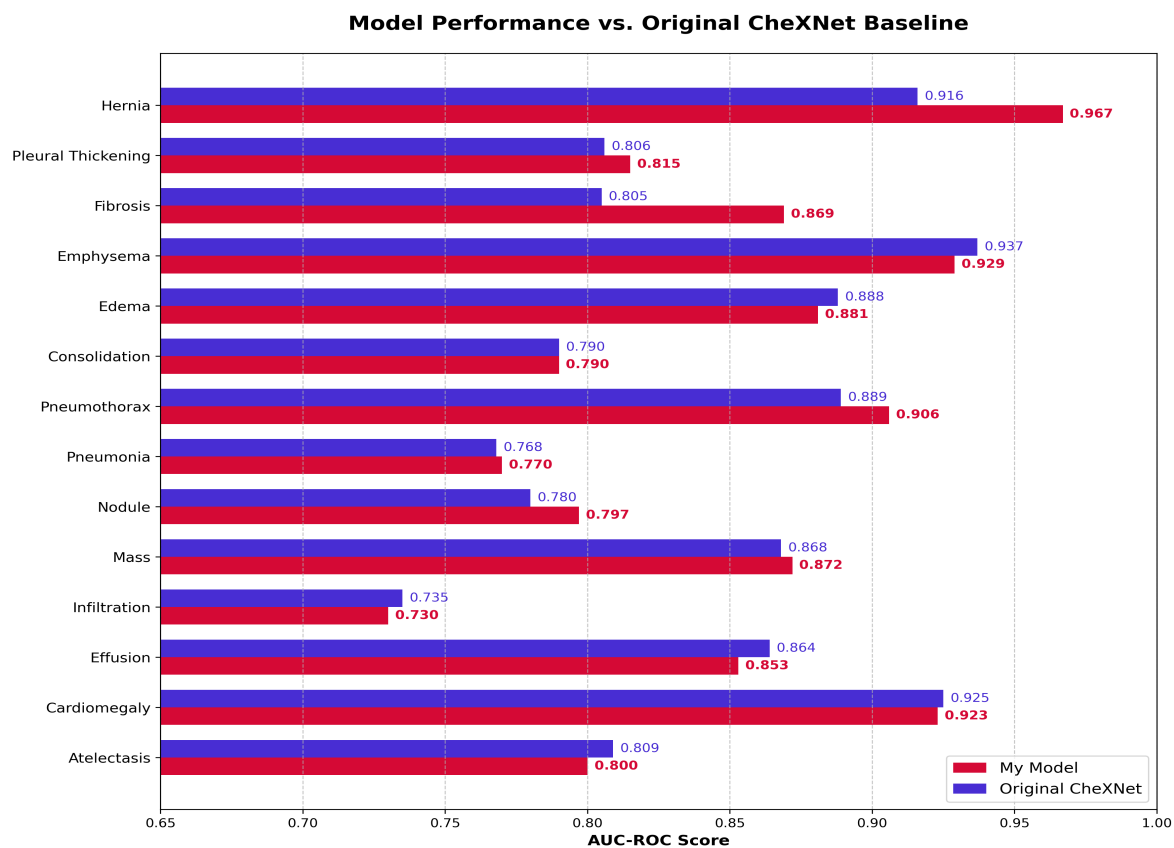


Figure 1 — AUC-ROC comparison per pathology class: this implementation (red) vs. original CheXNet baseline (blue). This model surpasses the baseline on 7 of 14 classes.

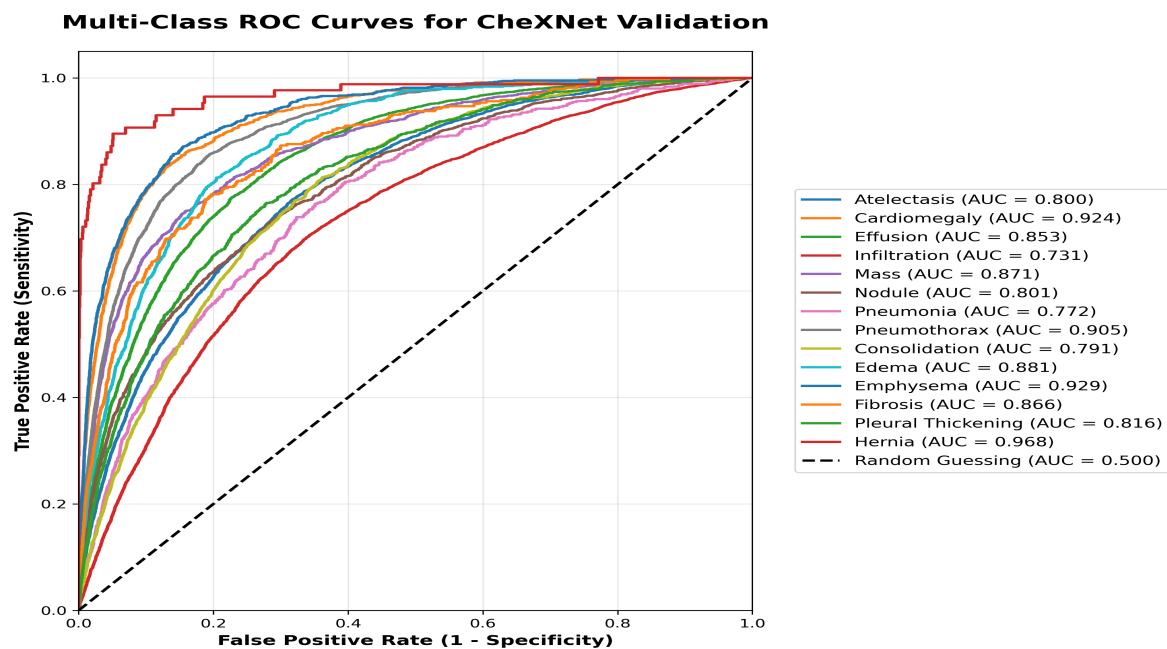


Figure 2 — ROC curves for all 14 pathology classes on the NIH ChestX-ray14 test set. Each curve corresponds to one disease; the dashed diagonal represents random chance (AUC = 0.500).

6. Explainability — Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) provides visual explanations for the model's predictions by producing a coarse heatmap that highlights the image regions most influential to a specific disease classification.

Method

For a target class c , gradients of the class score y^c are computed with respect to the feature maps A^k of the final convolutional layer (DenseBlock4, denselayer16.conv2). These gradients are global-average-pooled to obtain neuron importance weights α_k^c , which then weight and sum the activation maps. A ReLU is applied to retain only positive influences:

The resulting heatmap is resized to 224x224, normalised to [0,1], and overlaid on the original X-ray using a JET colormap at 40% opacity, producing an interpretable disease-localization visualization.

Pneumonia Localization Example

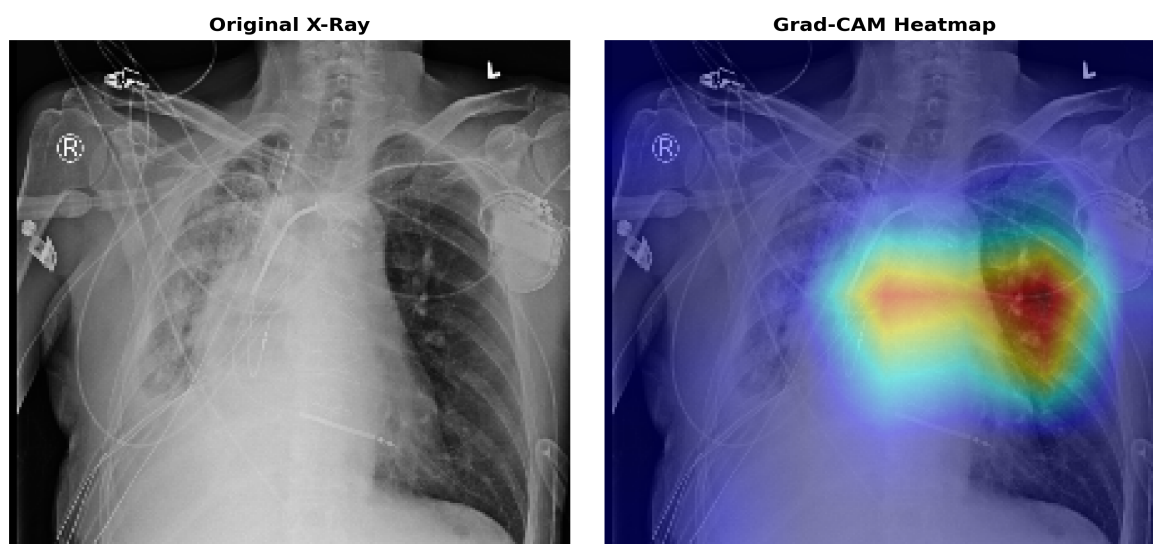


Figure 3 — Grad-CAM visualization for Pneumonia. Left: original chest X-ray. Right: Grad-CAM heatmap overlay. The model correctly focuses attention on the left lower lung field, consistent with clinical presentation of pneumonia.

The heatmap clearly demonstrates that the model attends to the anatomically correct region — the left lower lobe of the lung — rather than spurious artifacts such as medical devices, positioning markers, or image borders. This provides strong evidence that the learned representations are clinically meaningful.